

Azure Local Cloud Deployment



Agenda

- **Objective:** Step-through the Azure Local deployment process
- **Scope:** Prerequisites, hardware, networking, identity, security, and methodology (the Azure portal and ARM) with focus on hyperconverged architecture model



Deployment Prerequisites Overview

- **Prerequisite Categories:**
 - Azure prerequisites (subscription, permissions, regions)
 - System requirements (CPU, memory, NICs, storage)
 - Environment readiness (hardware, firmware, SBE package)
 - Physical & host networking requirements (DCB, VLAN, MTU, RDMA)
 - Firewall requirements (Outbound to Azure, internal ports)
 - Security prerequisites (TPM, Secure Boot, driver validation)
 - Active Directory requirements (organizational unit, LCM user account)

Azure Prerequisites (Subscription, Permissions, Regions)

- **Azure Subscription**
 - Supports Pay-as-you-go, EA, CSP, Visual Studio, trial accounts
- **Required Azure Permissions**
 - Contributor + User Access Administrator
 - Additional: Key Vault Contributor, Secrets Officer, Storage Contributor
- **Supported Azure Regions**
 - **Azure public:** East US, South Central US, Canada Central, West Europe, Australia East, Southeast Asia, India Central,, Japan East
 - **Azure Government:** US Gov Virginia
- **Purpose**
 - Required for activation, Azure Arc, update delivery, deployment orchestration

Hardware Requirements (Hyperconverged)

- **Node Count:** 1–16 identical machines
- **CPU:** Intel Nehalem / AMD EPYC, SLAT-enabled
- **Memory:** 32 GB ECC minimum
- **Boot Drive:** 200 GB minimum, 400 GB+ recommended
- **Data Drives:** At least 2× 500 GB (identical model, count, firmware) with S2D
- **Security:** TPM 2.0 + Secure Boot mandatory

Hardware Requirements (Disaggregated)

- **Compute Hosts:**
 - CPU, RAM, and NIC requirements similar to hyperconverged
 - No local S2D requirement
- **Storage Hosts:**
 - Require SAN HBAs, MPIO, consistent LUN presentation
 - FC-only SAN support (no iSCSI)
- **Homogeneity Rules:**
 - Compute hosts must be identical
 - Storage hosts must be identical
 - But compute $\neq$ storage hardware

Storage Prerequisites (Hyperconverged)

- **Storage Spaces Direct** dependencies:
 - Identical drive models, identical counts, identical firmware
 - NVMe, SSD, or HDD supported
- **Resiliency Models:**
 - 2-way mirror (1–2 nodes)
 - 3-way mirror (3+ nodes)
- **Bandwidth Requirements:**
 - High-bandwidth RDMA NICs for S2D
 - Separate **SMB01/SMB02** subnets recommended (VLAN 711/712)

Storage Prerequisites (Disaggregated)

- **External SAN Requirements:**
 - Fibre Channel only supported
 - All LUNs must be NTFS formatted
 - Each LUN must be dedicated to a single CSV
- **Host Requirements:**
 - Identical HBA configuration
 - MPIO must be installed and configured
- **Storage Network Characteristics:**
 - LUNs must be visible to all cluster nodes
 - SAN fabrics must be redundant

Networking Requirements (General)

- **NIC Requirements**
 - At least 2 NICs, certified in Windows Server Catalog
 - NICs must match **model**, **firmware**, **speed**
- **Switch Requirements**
 - Supports IEEE 802.1Q, 802.1Qbb, 802.1Qaz, LLDP
 - Must support **VLAN tagging**, **PFC**, **ETS** for RDMA
- **Traffic Types**
 - **Management**, **Compute**, **Storage**
 - Network ATC configures **intents** for each traffic class
- **Bandwidth**
 - 10 Mbit minimum to Azure services
 - High-speed $\geq$ 10/25/40/100 GbE recommended for storage

Networking Prerequisites (Hyperconverged)

- **Switch Requirements:**
 - Support for VLANs (802.1Q)
 - Support for PFC (802.1Qbb)
 - Support for ETS (802.1Qaz)
 - Support for LLDP (802.1AB)
- **NIC Requirements:**
 - NICs must match **vendor, model, speed, settings**
 - Must support **RDMA** for storage
- **Traffic Types:**
 - Management, Compute, Storage (SMB)
- **Network ATC:**
 - Provides **automatic intent-based configuration**

Networking Requirements (Disaggregated)

- **SAN Networking Considerations:**
 - Fibre Channel only (iSCSI not supported)
 - Must configure MPIO
 - All LUNs must be presented to all storage nodes
- **Compute Node Networking:**
 - Similar to hyperconverged except no S2D storage NICs
- **Switch Configuration:**
 - SAN switches must use consistent zoning, redundant fabrics, identical HBAs

Physical Networking Requirements & Traffic Models

- **North–South Traffic:**
 - Flows in/out of rack
 - Used for management, VM external access, multi-site traffic
- **East–West Traffic:**
 - Stays within rack
 - Used for S2D, Live Migration, storage replication
- **Switchless Option:**
 - Supported up to 4 nodes, storage-only
- **Topologies:**
 - 2-tier (Spine–Leaf)
 - 3-tier (Core–Aggregation–Access)

Host Network Requirements

- **Traffic Types:**
 - Management, Compute, Storage
- **NIC Qualification:**
 - NICs must support their assigned roles:
 - Management
 - Compute Standard/Premium
 - Storage Standard/Premium
- **SET (Switch Embedded Teaming):**
 - Required for **teaming** of adapters
 - Supports RDMA, Dynamic VMMQ

Firewall & Security Requirements

- **Outbound Requirements**
 - Allow ports 80/443 to Azure
 - No TLS inspection permitted
- **Internal Cluster Ports**
 - SMB 445, SMB Direct 5445
 - WinRM 5985/5986
 - RPC 135, dynamic RPC ports
 - ICMP, NTP 123
- **Service Tags**
 - Use Azure service tags for dynamic allowlists
- **OEM Endpoints**
 - Additional URLs required for DataON, Dell, HPE, etc.

Azure Connectivity Requirements

- **Outbound Traffic:**
 - Must reach Azure over HTTP/HTTPS
 - Required for:
 - Azure Arc onboarding
 - Update retrieval
 - Azure Resource Bridge
 - Deployment sequencing
- **Unsupported:**
 - TLS inspection
 - Private Link Scopes for Arc
- **Tools:**
 - Use **Environment Checker** to validate outbound connectivity

Active Directory Requirements

- **AD Requirements:**
 - Dedicated **Organizational Unit (OU)**
 - OU path should not include: & " ' < >
 - **Blocked GPO inheritance** at OU level
 - A new **LCM deployment user** with required rights
- **Machine Requirements:**
 - Machines must **not** be domain-joined before deployment
- **Purpose:**
 - Provide a designated OU for **cluster objects, deployment user, and machine accounts**

Active Directory Preparation Commands

- The `AsHciADArtifactsPreCreationTool` module automates AD prep
- Key cmdlet: `New-HciAdObjectsPreCreation`
- Required parameters:
 - `AzureStackLCMUserCredential`
 - `AsHciOUName` (Distinguished Name format)
- Password Requirements:
 - `≥14 characters`, with lowercase, uppercase, numeral, special character
- Outcome:
 - New OU,
 - New LCM deployment user,
 - Blocked GPO inheritance

Verifying Identity Integration

- **Active Directory Checks:**
 - Nodes appear in the correct OU
 - LCM user part of local administrators
 - No GPO interference
- **Key Vault Identity Checks:**
 - Secrets visible in Key Vault
 - RecoveryAdmin credential exists
 - Vault access policies configured

Downloading the Azure Local Operating System

- **Source:**
 - Download via the **Azure portal** under **Azure Local** → **Get Started** → **Download Software**
- **Requirements:**
 - Azure account with the **Reader** role or higher
 - **Azure Stack HCI resource provider** registered
- **Supported versions:**
 - Portal deploys only **recommended versions**
 - Older builds require **ARM template deployment**
- **Notes:**
 - Language support: **English only**

OS Installation Workflow

- **Key Steps:**
 - Boot from ISO
 - Select **English (US)**
 - Accept licensing
 - Choose **Custom: Install newer version (advanced)**
 - Select and prepare OS drive
 - Restart sequence completes installation
- **Post-Install:**
 - Set **local administrator password** (14+ chars, complexity)
 - Confirm OS boots to **Sconfig**
 - Configure **HBA drivers, MPIO, and SAN connectivity (hyperconverged)**

Configuring the OS Using SConfig

- **Key SConfig options:**
 - Network Settings (IP, DNS, VLAN)
 - Computer Name
 - Remote Desktop (optional)
 - Windows Update (do NOT use — updates installed during deployment)
 - Domain Join (only if using v2604+ “pre-join” support)
- **Critical Settings:**
 - DNS must point to **AD domain DNS servers**
 - Time sync must be configured via **w32tm**

Domain Join Before Deployment (Optional Feature)

- Supported starting with **version 2604**
- **Requirements:**
 - Add LCM deployment user to local Administrators
 - Join using SConfig → Option 1 (Domain/Workgroup)
- **Benefits:**
 - Simplifies cluster creation in AD-centric environments
- **Caution:**
 - Must apply identical local admin credentials across nodes

Configuring Drivers and Firmware (Hyperconverged)

- **Required Actions:**
 - Update all NIC, chipset, NVMe/SSD, HBA, and GPU firmware
 - Install vendor-provided Solution Builder Extension (SBE) under C:\SBE
- **Driver Validation Steps:**
 - Identify NICs using `Get-NetAdapter`
 - Verify driver details using `Select-Object *driver*`
 - Avoid inbox drivers (`DriverProvider = "Microsoft"`)

Configuring Drivers and Firmware (Disaggregated)

- **Storage Nodes:**
 - Update HBA drivers, firmware, and SAN provider add-ons
 - Configure MPIO
- **Compute Nodes:**
 - Update NIC drivers, chipset, and management firmware
- **Validation:**
 - Use `Get-NetAdapter` for NICs
 - Use vendor tools to validate SAN HBA configuration

Azure Arc Registration (Without Gateway)

- **Two registration methods:**
 - Arc script (PowerShell)
 - Configurator App (UI-based workflow)
- **Required Parameters:**
 - Tenant ID, Subscription ID, Resource Group, Region
- **Registration Script:**
 - `Invoke-AzStackHciArcInitialization`
- **Requirements:**
 - Machine must run **release 2505+**
 - Must have outbound HTTPS access
 - LCM user must have correct Azure RBAC roles

Azure Arc Registration (With Gateway)

- Uses an on-premises Arc Gateway to proxy outbound traffic
- **Requirements:**
 - Machines running **release 2506+**
 - **ArcGatewayID** parameter required
 - Gateway must be deployed in same subscription
- **Validation:**
 - Check **arcproxy.log**
 - Use **azcmagent show** and **azcmagent check**
- **Use cases:**
 - Enterprise requires **proxy control**, **centralized routing**, or **tightly controlled outbound traffic**

Validating Arc Registration & Remediation

- **Validation Commands:**
 - `Get-ArcBootstrapStatus`
 - `Collect-ArcBootstrapSupportLogs`
- **Status States:**
 - **Succeeded** - ready for deployment
 - **Update: InProgress** - OS baseline is being updated
 - **Failed** - requires script re-execution
- **Remediation Steps:**
 - Re-run **`Invoke-AzStackHciArcInitialization`**
 - Re-authenticate with new device code
 - Validate outbound connectivity and identity roles

Deployment via Azure Portal

- **Deployment Flow:**
 - Create Azure Local instance in Azure portal
 - Add Arc-registered machines
 - Select identity model (AD or Key Vault)
 - Configure networking, management, security, advanced
 - Run validation, then deploy
- **Tools Involved:**
 - Azure Portal
 - Azure Resource Bridge
 - Azure Arc agents
- **Goal:**
 - Initiate cluster build

Portal Deployment: Adding Machines

- Machines must appear as Arc-enabled servers
- Requirements:
 - All machines must show status Ready
 - Must run the same OS version
 - Must have identical hardware profiles
- Validation Step:
 - Run Validate selected machines
 - Portal checks OS version, extensions, NIC symmetry, SBE availability

Portal Deployment: Identity Settings (Active Directory)

- **Provide:**
 - AD Domain FQDN
 - OU Distinguished Name
 - LCM Deployment User credentials
- **Requirements:**
 - OU must already exist
 - GPO inheritance must be **blocked**
 - LCM user's password must meet **complexity requirements**

Portal Deployment: Identity Settings (Key Vault)

- **Local Identity option stores:**
 - Local admin credentials
 - BitLocker keys
 - Cluster secrets
- **Requires:**
 - A new or existing **Azure Key Vault**
 - Vault must allow **public network access**
 - User must have **Key Vault Secrets Officer + Key Vault Contributor**
- **Goal:**
 - Replace AD requirement with **certificate-based** identity

Portal Deployment: Networking (Hyperconverged)

- **Must define network intents:**
 - Management
 - Compute
 - Storage
- **Select appropriate NICs for each intent**
- **Required VLANs:**
 - Storage NICs typically use VLAN 711/712
- **IP Assignment:**
 - Management can be **DHCP** or **static**
 - Storage is **automatic** (ATC-assigned)

Portal Deployment: Networking (Disaggregated)

- **Compute nodes:**
 - Configure **Management** and **Compute** intents
 - No S2D storage intent
- **Storage nodes:**
 - Configure **SAN connectivity** externally
 - Portal does not configure storage NICs
- **IP Assignment:**
 - **Management** intent required
 - No storage IPs auto-assigned

Portal Deployment: Management Settings

- **Custom Location Name:**
 - logical container for resource placement
- **Cloud Witness Storage Account:**
 - Required for 2-node systems
 - Automatically configured
- **Local Admin Credentials:**
 - Must be identical on all nodes
 - Must meet complexity requirements
- **DNS Server:**
 - Must not belong to Kubernetes reserved ranges

Quorum Configuration & Witness Setup

- **Quorum Types:**
 - Cloud Witness (recommended for 2-node clusters)
 - Node Majority (3+ nodes)
- **Portal automatically configures the chosen witness type**
- **Cloud Witness Requirements:**
 - Dedicated **Storage Account** per cluster
 - Azure connectivity (Outbound TCP 443)

Portal Deployment: Security Settings

- **Recommended Security Baseline** enables:
 - Credential Guard
 - SMB Signing
 - SMB Encryption
 - BitLocker on OS & data volumes
 - WDAC (Application Control)
 - Drift Control
- **Customized Mode** allows disabling specific controls

Portal Deployment: Advanced Settings

- **Volume Creation Mode:**
 - **Express** — create data volumes (one per node) in addition to infra volume
 - **Infrastructure Only** — create just the cluster infra volume
 - **Keep Existing Storage** — reuse Storage Spaces pool (single node)
- **Tags:**
 - Can apply Azure tags to cluster resource
- **SBE Version:**
 - Select vendor-provided build if applicable

Deployment Validation Phase

- **Validates:**
 - Hardware compliance
 - Driver and firmware versions
 - Network intent configuration
 - Identity and Key Vault access
 - Storage configuration
 - Outbound Azure endpoints
- **Validation Time:**
 - ~15 minutes for 1–2 nodes
 - Longer for large clusters

Deployment Sequence Overview

- **Azure Local deployment proceeds through sequential phases:**
 - Environment validation
 - Identity configuration (AD or Key Vault)
 - Network intent creation
 - Cluster creation & Arc Resource Bridge deployment
 - Storage initialization
 - Workload readiness validation
- **Automation:**
 - Azure portal orchestrates all steps end-to-end via Azure Arc and internal deployment agents.

Hyperconverged Deployment Sequence

1. Prepare Active Directory
2. Arc-register all nodes
3. Run portal validation
4. Apply Network ATC intents
5. Create cluster & assign quorum
6. Enable Storage Spaces Direct
7. Deploy infrastructure volume & workload volumes

Disaggregated Deployment Sequence

1. Arc-register compute and storage hosts
2. Run portal validation
3. Configure compute networking (no S2D storage intents)
4. Cluster creation on compute nodes
5. SAN integration:
 - Configure MPIO
 - Validate Fibre Channel zoning
 - Mount LUNs as Cluster Shared Volumes
6. Initialize workload volumes

Azure Resource Bridge (ARB) Deployment

- **Purpose:**
 - Enables Azure Arc-based VM provisioning
 - Provides custom location, infrastructure network, and resource projection
- **Key Components:**
 - Kubernetes-based ARB Stack
 - Management endpoint
 - Local infrastructure VM
- **Deployment Notes:**
 - ARB is deployed automatically
 - Requires infrastructure logical network (static range)

Monitoring Deployment Progress

- Deployment progress visible under **Azure Local → Deployments**
- **Key stages:**
 - Initialize environment
 - Configure identity
 - Deploy ARB Stack
 - Create cluster
 - Set up volumes
- **Expected timelines:**
 - Single node: ~1.5–2 hours
 - Two-node system: ~2.5 hours
 - Larger clusters scale accordingly

Post-Deployment Tasks (Hyperconverged)

- Verify S2D health via cluster validation
- Confirm volumes, storage pool, tiering, resiliency
- Configure network ATC drift control
- Optional:
 - Enable RDP
 - Configure Log Analytics
 - Enable Azure Monitor Alerts

Post-Deployment Tasks (Disaggregated)

- Validate SAN connectivity and MPIO across hosts
- Confirm LUNs are mapped as Cluster Shared Volumes
- Validate heartbeat traffic, cluster roles, and VM placement
- Optional:
 - Configure SAN vendor monitoring
 - Enable Azure alerts for storage access failures

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